CDL Permit & Endorsement Prep: Build Spec v2 (lean MVP + stage gates)

For: Claude Code (`brief` agent), with Sean as owner and reviewer. **Written:** Wed Oct 28, 2026, PT. **Replaces:** `cdl-build-spec-2026-10-28.md` (v1, kept unchanged).

Folds in: - the §14 diff in `cdl-ux-differentiation-2026-10-28.md`; - the positioning and moat stack in `cdl-moat-2026-10-28.md`; - the factory rule Sean confirmed: many lines in the water, rotate, invest only in the handful of winners.

Marking: - [VERIFY] = confirm from the primary source before shipping. - [E] = estimate. [J] = judgment. - **[LEAN-MOVE]** = can move after launch to shorten the MVP (§13).

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0. What changed from v1 (summary)

Area	v1	v2
Positioning	"$34.98 once. Everything. Forever. No ads. No auto-renew."	Leads with exactness and proof: "The CDL test prep that matches your state's real test, fixed in public when the rules change." Price line is second (§1)
Design	Not specified	5 design principles, brand direction, large type, Sun Mode and accessibility are P0 (§1, §8)
Practice feedback	Inline explanation	**Proof sheet** with a source card (paraphrased excerpt + official manual link) (§5)
Results	Score first	Leads with **Replay mistakes**, default filter "Missed"
New screen	—	**Test Day plan** (countdown, Ready check, exam-day card) [LEAN-MOVE]
Paywall	From any locked item	Only at a section boundary **after the free test completes**; never mid-test; no trial; guardrail metric = 0 mid-test views
Onboarding	3 steps	Max 3 screens, searchable state list, "Not sure — use national", **no-loop QA test**
Listen mode	P1	Renamed **Cab Mode** (P1), with safety line; lock-screen audio [LEAN-MOVE]
Moats in MVP	Source chip, fix log	**Exact-format engine** (TX/CA/FL + national at MVP; data model ready for 51 jurisdictions), manual citations, **rule-change watch + public fix log** (week 3 if it fits, else days 22–35)
Content lint	6 checks	+ reading grade ≤ 8, grammar pass, figure check, section integrity, state scoping, HazMat trap regression
Post-MVP	Calendar (days 22–35)	**Stage gates tied to sales** (Gate 0–3), kill lines kept (§13)
Scope	21 days	Full v2 = 21 days. **Lean MVP ≈ 15 days [E]** (§12)
New	—	**§16 Factory template**: reusable modules for the next licensing-exam apps

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1. Product summary, positioning and design principles

What it is: a mobile CDL knowledge-test prep app. - **Permit (CLP):** General Knowledge, Air Brakes, Combination. - **Endorsements:** HazMat, Tanker, Doubles/Triples, Passenger, School Bus. - **States at MVP:** Texas (with the TX Commercial Rules test), California and Florida, plus a national fallback for every other state. - The data model is ready for all 51 jurisdictions (§6). They unlock at Gate 1.

Positioning (new, from the moat report):

> **"The CDL test prep that matches your state's real test, fixed in public when the rules change. Every endorsement, $34.98 once. No ads, no auto-renew."**

- App Store subtitle: "Your state's exact CDL test" (27 chars) [VERIFY final characters against the name].
- Price microcopy: "Pay once. Yours to keep. No ads. No auto-renew."

Three proof points (first screen and screenshots 1–3): 1. **Your state's real format.** Exact count, pass mark, answer choices and order, each with its source. For example: TX Commercial Rules is taken first; 20 questions, 80% [VERIFY TX DPS]. 2. **Every answer shows its source.** Manual section, CFR reference and "Last checked", plus a public fix log. 3. **Honest pricing.** One-time only, a free test in every section, Restore that works, no trial, never a mid-test paywall.

Design principles (every screen is checked against these): 1. **Never interrupt.** No mid-test paywalls, prompts, rating asks or sounds. 2. **Show the proof.** The source is one tap away on every answer. 3. **Never lose anything.** Autosave on every answer; resume exactly. 4. **Readable anywhere.** Large type, glare-proof Sun Mode, no truncation, one-handed. 5. **Plain English.** About an 8th-grade reading level or lower, with terms defined inline.

Brand direction: - **Feel:** calm, factual, built for professionals, "a good instructor who tells you straight." Not a game, not hype. - **Type:** Atkinson Hyperlegible, or Inter if it renders better in React Native. Body text at least 18pt, scaling to accessibility sizes. Tabular numbers for scores. - **Color:** charcoal on off-white with one road-sign accent (highway green; amber as accent only). Right and wrong always use color + icon + word. **Never use hazmat placard colors as decoration.** Dark mode follows the system. Sun Mode is maximum contrast with thicker strokes. - **Imagery:** line icons; original diagrams only where the manual has one (air brakes, placards). No mascots, laurels or "PASSED" stamps. - **Never say:** "pass first try", "real test questions", a pass rate, "Official", "DMV" or "DPS" in the name. - **Motion and sound:** subtle. No autoplay sound. Haptics on answer, with a toggle.

Pricing (unchanged; approved):

Product	Price	Unlocks
Free	$0	One practice test per section, with full Proof sheets, citations and audio. No ads, ever.
Permit Pack	$14.99 once	GK, Air Brakes, Combination for every state (TX adds Commercial Rules); full bank, simulator, Replay mistakes, readiness
Endorsement (each)	$4.99 once	One of the 5 endorsements
All Endorsements	$19.99 once	All 5, including modes added later
Permit + All	$34.98	Permit Pack + All Endorsements

- **Rules:** no subscriptions, trials, ads SDK or rewarded ratings.
- **Price test:** $19.99 Permit Pack only at Gate 3.
- **[VERIFY]:** store price points exist. If "Everything" is ever a single SKU, use the nearest point ($34.99).

2. Target users (unchanged from v1)

1. First-time CLP applicants, in or near ELDT (about 390k a year [E]).
2. CDL holders adding an endorsement or renewing HazMat (endorsement-only mode; never pushed to Permit).
3. TX and FL test-takers facing English-only knowledge tests (FL since Feb 6, 2026; TX since Jun 1, 2026):
4. plain-English explanations, glossary and audio;
5. test content stays in English.
6. The Spanish "English-test bridge" waits for Gate 2.
7. Recruits referred by carriers or schools. This is organic only; no outreach.

3. Platform and stack (unchanged from v1)

- **App:** Expo (React Native, TypeScript), Expo Router.
- **Data and state:** `expo-sqlite` (read-only content DB + user DB); Zustand persisted to SQLite.
- **Purchases:** RevenueCat free tier or `react-native-iap` (decide Day 4).
- **Audio:** `expo-speech`.
- **Updates, analytics, crashes:** EAS Update for OTA content; PostHog anonymous; Sentry.
- **Backend and web:** Cloudflare Worker + D1 for mistake reports; Astro on Cloudflare Pages for the web.

- **CI and tests:** EAS Build, Jest, one Maestro flow.
- **Hard rule:** no ads, attribution or tracking SDK. Privacy label: "Data Not Used to Track."
- **New, for the rule-change watch:** a scheduled GitHub Action or Cloudflare Cron Trigger (free tier) (§7.8).

4. One-time IAP setup (unchanged from v1, plus stricter paywall rules)

Product IDs: - `permit_pack` $14.99 - `endo_hazmat` / `endo_tanker` / `endo_doubles` / `endo_passenger` / `endo_schoolbus` $4.99 each - `endo_all` $19.99

Store requirements: - Entitlements are checked at launch and on foreground (StoreKit 2 `currentEntitlements` + `Transaction.updates`). - Restore is always visible on the paywall and in Settings. - Play purchases are acknowledged within 3 days. - Family Sharing is on for iOS non-consumables. Don't promise it on Android [VERIFY]. - Refunds re-lock quietly. - Entitlements are cached for offline use. - Apple Small Business Program (15%).

Paywall rules (new, strict): 1. The paywall opens only: - (a) at a section boundary after the free test is **completed**; or - (b) when the user taps a locked item. It **never** opens mid-test, mid-drill or in Cab Mode. 2. Copy: live store prices, then "Pay once. Yours to keep. No ads. No auto-renew." The screen lists what's free and what each item unlocks. 3. No trial, no countdown timers, no "limited offer", no pre-checked bundle. 4. Bundle math: "All Endorsements $19.99 (you own N)". 5. Close (×) is always visible, at least 44pt, top-left or top-right, with no delay. 6. Endorsement-goal users never see the Permit Pack first. 7. The guardrail metric `paywall_viewed{mid_test:true}` must equal **0**. A CI or analytics alert fires if it's ever above 0.

5. Screens and user flows

Screens: 1. **Onboarding (max 3 screens, no account):** - (1) Searchable state list with TX/CA/FL pinned and "Not sure — use national". - (2) Goal: "Getting my permit", "Adding an endorsement" or "Renewing HazMat". - (3) "What's free" card, then straight to Home. - **QA test:** the state picker can't loop. Run Maestro on small and large Android screens and iPhone SE/Pro Max. 2. **Home:** section cards, each showing readiness, free or unlocked status, and "Continue". HazMat comes first for endorsement-goal users. On a state-specific track it shows a **format card**, e.g. "Texas: Commercial Rules (20 Q · 80%) → GK (50 · 80%) → …", with a source link. 3. **Section detail:** Free test, Practice by topic, Simulator, Replay mistakes, Cab Mode. 4. **Question screen:** - full-width answer rows that wrap, with no truncation at any size; - tap targets of at least 48pt; - a state badge on state-specific questions; - Read aloud and Flag buttons. 5. **Proof sheet** (practice mode, after each answer): - "Correct" or "Not quite", then the right answer in bold. - **Why** in 1–3 plain sentences. - Source chip, e.g. `TX CDL Handbook §9.6.2 · 49 CFR 172.202 · Last checked Oct 2026`. - Buttons: 🔊 Read aloud, Flag a problem, Continue. - Tapping the chip opens a **Source card**: a paraphrased excerpt (≤240 chars, never verbatim) plus a link to the free official manual. - Misses go to the Replay queue automatically. 6. **Simulator:** - the verified state format (count, pass mark, answer choices, time limit only if published); - no feedback until the end, "like the DMV." 7. **Results:** - leads with **"Replay N mistakes"** as the primary CTA; - the filter defaults to Missed; - then score, pass/fail and topic breakdown. 8. **Replay mistakes:** spaced-repetition queue (Leitner, 5 boxes: 1/2/4/7/14 days). 9. **Test Day plan** [LEAN-MOVE → v1.0.1]: - optional test date and countdown ring; - **Ready check** per section ("Seen 92% · Mastered 78% · Last 3 sims 84/88/90"); - "Two topics holding you back"; - today's plan in minutes; - the night before: "Replay all mistakes" plus an **Exam-day card** (state format from StateFormat; "what to bring" only once sourced [VERIFY]). 10. **Cab Mode** (was Listen mode): - big type, Sun Mode, 2 buttons; - reads question, pause, options, pause, answer and why; - speed 0.75–1.5×; works offline; - safety line at start: "Listen only — don't use the screen while driving"; - lock-screen and background audio [VERIFY] [LEAN-MOVE]; - voice answers come at Gate 3. 11. **Glossary:** plain-English and searchable; tap-to-define in questions; optional Spanish gloss on terms only. 12. **Paywall:** per the §4 rules. 13. **Settings:** state, goal, text size, Sun Mode, voice and speed, Restore, "What's new and fixed" (fix log), privacy, support. 14. **Report a mistake:** - reasons: Wrong answer / Outdated / Not in my state's manual / Confusing wording / Typo / Other; - optional note; question ID attached automatically; - confirmation: "Thanks — we check every report. Fixes appear in What's new."

Key flows: - **First run:** state → goal → Home → free GK test → Results ("Replay N mistakes") → then a soft paywall card at the section boundary. - **Purchase** returns the user to the exact place. **Restore** needs no account. - **Endorsement-only:** the HazMat free test, then the $4.99 / $19.99 paywall. Permit is never required. - **Autosave** on every answer. Interruptions resume. A back swipe asks for confirmation. - **Rating prompt:**

native API, at most once, **only on Home** after the 2nd passed simulator. Never in a session and never tied to a reward.

6. Data model (v1 plus new fields)

```
type Jurisdiction = 'NAT'|'AL'|'AK'|...|'WY'|'DC';  // all 51 + NAT defined now; only NAT/TX/CA/FL populated at MVP
type Section = 'GK'|'AIR_BRAKES'|'COMBINATION'|'TX_COMMERCIAL_RULES'|'HAZMAT'|'TANKER'|'DOUBLES_TRIPLES'|'PASSENGER'|'SCHOOL_BUS'
  | 'HAZMAT_RENEWAL'|'AIR_BRAKE_RESTRICTION'; // last two reserved for Gate 1
interface Question {
  id: string; state: Jurisdiction; section: Section; topic: string;
  question: string; options: string[]; answer_index: number; explanation: string;
  manual_citation: string; manual_edition: string; regulation_ref?: string;   // e.g. "49 CFR 172.504"
  cfr_sections: string[];        // NEW: machine-readable, e.g. ["49/172/504"]; the rule-change watch keys on these
  source_a: {doc:string; locator:string; url?:string}; source_b: {doc:string; locator:string; url?:string};
  source_excerpt_paraphrase: string;  // NEW: <=240 chars, never verbatim (CI check)
  official_manual_url: string;        // NEW
  figure_asset?: string;              // NEW: lint fails if the text says "shown/figure/diagram" without one
  reading_grade: number;              // NEW: computed in CI
  verified_a_by: string; verified_b_by: string; last_checked: string;
  review_status: 'draft'|'agent_verified'|'human_verified'|'published'|'flagged'|'retired';
  difficulty: 1|2|3; glossary_terms: string[]; free_tier: boolean; version: number; change_note?: string;
}
interface StateFormat {             // the exact-format engine; one row per jurisdiction x section
  state: Jurisdiction; section: Section; required_for: ('A'|'B'|'C'|'ENDO')[];
  order: number;                     // e.g. TX Commercial Rules = 1
  question_count: number; pass_pct: number; options_per_q: 3|4; time_limit_min?: number;
  test_language: 'en_only'|'en_plus'; languages?: string[];   // FL/TX = en_only (2026)
  manual_edition: string; manual_model: 'legacy'|'aamva_modernized'|'unknown';
  source_url: string; last_checked: string; verified_by: string; notes?: string;
}
interface WatchedSource { id: string; kind: 'ecfr_part'|'fed_register_query'|'manual_url'|'dmv_page';
  target: string; last_hash: string; last_seen: string; }
interface FixLogEntry { date: string; question_id?: string; state?: Jurisdiction; version: number;
  kind: 'user_report'|'rule_change'|'manual_edition'|'format_change'; summary: string; source_url?: string; }
interface GlossaryTerm { term: string; plain: string; es_gloss?: string; citation?: string; }
```

- **User DB:** unchanged from v1 (attempts, srs, sims, progress, entitlements, pending_reports), plus `test_date` and `settings(text_size, sun_mode, haptics)`.
- **Simulator defaults** (national fallback): GK 50 / Air Brakes 25 / Combination 20 / HazMat 30 / Tanker, Doubles, Passenger, School Bus 20 each, all 80% [VERIFY each state].
- **Known format facts to load** (from the moat report; [VERIFY] each one on Day 1–2):
- TX: Commercial Rules 20 Q / 80%, taken first. A third-party site says 25, so resolve against TX DPS.
- CA: GK 50 Q, 10 misses allowed.
- FL and TX: English-only.
- NY and IA: 3 answer choices (Gate 1 rows).
- Ship nothing in the simulator that StateFormat hasn't verified.

7. Content pipeline

Sources (unchanged): - **Allowed:** 49 CFR (public domain), FMCSA guidance, and state manuals paraphrased with section citations. - **AAMVA:** the model manual is copyrighted; paraphrase only. - **Forbidden:** competitor banks, leaked or "actual test" questions, scraping competitors.

Steps 1–4 (unchanged): topic map → agent draft → agent Check 1 (fresh context, source_b) → human Check 2. - Human Check 2 covers 100% of HazMat and TX Commercial Rules, every flagged item, and a 20% sample of the rest. - A batch goes back if the sample error rate is over 2%.

Step 5, CI lint (v1 checks plus new): - **v1 checks:** - exactly one correct answer; - no unintended "all/none of the above"; - options of similar length; - citation and both sources present, `last_checked` ≤ 12 months; - no 12+ word verbatim match against source PDFs; - near-duplicate check (embedding similarity > 0.92). - **New checks:** - Explanation reading grade ≤ 8 (Flesch-Kincaid). - Grammar and spelling pass (LanguageTool) on question, options and explanation. - `source_excerpt_paraphrase` ≤ 240 chars and fails the verbatim check against the manual. - Figure check: references to a figure require `figure_asset`. - Section integrity: `section` matches the topic map (no bus questions in Combination). - State scoping: non-NAT questions are only drawn for their state. - No repeated IDs or near-duplicates within one simulator draw. - **Known-trap regression tests**, e.g. HazMat basic-description order (ID → proper shipping name → class → packing group, 49 CFR 172.202 [VERIFY]). - Every published question has non-empty `cfr_sections` or a manual citation that maps to a WatchedSource.

Step 6, Report a mistake (unchanged): - reports go to D1, with a daily digest to Sean; - triage within 72h; - a fix bumps `version`, writes a FixLogEntry and ships OTA; - 3+ reports auto-hide the question.

Step 7, re-check: quarterly, plus whatever the watch flags.

Step 8, rule-change watch (new; moat #2): - **What it does:** a daily job (GitHub Action or Cloudflare Cron) runs three checks and writes a digest to Sean: 1. **eCFR versioner API** (`/api/versioner/v1/versions/title-49.json?part=N`) for Parts 383, 391, 392, 393, 395, 396, 397 and 172. A new amendment date flags every question whose `cfr_sections` sits in that part. 2. **Federal Register API** for PHMSA and FMCSA rules and NPRMs (e.g. HM-215S, comments due Dec 28, 2026). These are logged as "watching", with no question change until final. 3. **Hash of each state manual and DMV CDL page** (TX DPS, CA DMV, FL FLHSMV at MVP). A change flags that state's StateFormat rows and questions. - **Output:** flagged items go to `review_status: 'flagged'`. A fix ships OTA with a `rule_change` FixLogEntry. - **Public log:** `/fixes` lists every entry with date, what changed and the source link, both in-app and on the web. - **Why it matters:** in 2026 so far, Part 383 has had 8 amendment dates and Part 172 has had 7 (moat report). That record compounds and can't be backfilled. - **Placement:** MVP days 18–20 if it fits. Otherwise Gate 0+, days 22–35. It's a separate job with no app-review dependency.

Launch content targets:

Section	v1	Lean v2 [E]	Note
General Knowledge	300	250	5 non-repeating 50-Q sims
Air Brakes	120	100	
Combination	100	80	
TX Commercial Rules	80	60	3 sims; 100% human-checked
HazMat	200	160	100% human-checked
Tanker / Doubles / Passenger / School Bus	80 each	50 each	2.5 sims each. Grows OTA to 80 at Gate 1 [LEAN-MOVE]
Total	~1,120	**~850**	Cuts Sean's review from about 13h to about 10h [E]

- Free tier: 1 set per section (20 Q, or 30 for GK).

8. Features, prioritized (v2)

P	Feature	Done means
P0	Offline question engine with autosave	Airplane mode works; killing the app mid-test resumes on the same question
P0	**Exact-format engine** (StateFormat for TX/CA/FL + NAT; 51-jurisdiction schema) and format card	Choosing a state changes the bank, simulator and order; every row has source + last_checked
P0	Proof sheet + source chip + Last checked + Source card	100% of published questions; CI fails otherwise
P0	Free test per section; strict paywall rules (§4)	Reachable with no prompt; 0 mid-test paywall views in QA
P0	One-time IAP + Restore + Family Sharing (iOS)	Buy → delete → reinstall → Restore works on both stores

P	Feature	Done means
P0	Simulator in the state's format	Matches verified StateFormat
P0	Results leading with **Replay mistakes** + Leitner drill	Results CTA = Replay; queue persists
P0	HazMat first; TX Commercial Rules	HazMat and TX CR 100% human-checked
P0	**Large type, Dynamic Type to accessibility sizes, Sun Mode, no truncation**	Snapshot test at the largest size passes on key screens
P0	**VoiceOver/TalkBack** on question, Proof sheet and results	Manual audit passes
P0	3-screen onboarding, no-loop	Maestro no-loop test passes on 4 screen sizes
P1	Cab Mode (foreground audio)	TTS sequence, speed control, offline, safety line
P1	Report a mistake + public fix log	A report reaches D1 within 1 minute online; fix log renders
P1	Readiness score on Home and Results	Formula as v1; "Likely ready" ≥ 85; "a study guide, not a guarantee"
P1	Glossary + tap-to-define	**80 terms at MVP** (v1: 150) [LEAN-MOVE the rest]
P1	Rule-change watch (§7.8)	Daily digest runs; flags propagate
P1 → v1.0.1	Test Day plan + Ready check breakdown	[LEAN-MOVE]
P1 → v1.0.1	Cab Mode lock-screen/background audio	[LEAN-MOVE] [VERIFY]
P2	Endorsement-only mode (hide permit upsells)	Goal = endorsement hides Permit cards
P2	Dark mode	Follows the system

- **Readiness formula** (unchanged): `0.5×mastered + 0.3×avg(last 3 sims) + 0.2×seen`. Never claim a pass rate.
- **Out of scope until a gate unlocks it:**
- 48 more states, SEO beyond the seed pages, endorsement modes (Gate 1).
- Miss-rate pages, Spanish bridge, pre-trip audio, video pipeline (Gate 2).
- Voice answers, price test, motorcycle, paid examiner review (Gate 3).
- **Never:** subscriptions, ads, accounts, live AI chat, a pass guarantee (Apple controls IAP refunds), leaderboards, a community forum.
- **Roadside English** (v1 fast-follow) moves into the Gate 2 Spanish bridge.

9. Store listing, ASO and naming

- **Naming rules** (unchanged):
- no "Official/DMV/DPS/FMCSA" or seals;
- avoid "CDL Practice Test®" (PATG mark);
- Sean runs a USPTO search.
- **Names:** HaulReady, Straight Answers CDL, CleanRun CDL, Plainspoke CDL, CDL Proof, Checkride CDL.
- **iOS:** title "HaulReady: CDL Permit Test". Subtitle "Your state's exact CDL test". Keywords as v1, plus "audio" and "hazmat practice" [VERIFY character budget].
- **Play short description:** "Your state's exact CDL test. Every answer cited. Pay once, no ads."
- **Screenshots:**
- "Pay once. No ads. No auto-renew."
- "Every answer shows its source." (Proof sheet with the source card open)
- "Texas? Commercial Rules included. HazMat done right." (format card + HazMat placard question)
- Ready check / Test Day (use Results + readiness if Test Day is LEAN-MOVED)
- Cab Mode
- "See what we fixed" (fix log)
- **Screenshot visuals:** neutral background, one accent, large type, no mascots, laurels or "first try" claims.
- **Description:** add "readable", "audio", "no ads", "your state's format".
- **4.3 spam mitigation** (unchanged, and more important under the factory model; see §16): one app per exam, a distinct UI and original bank, detailed review notes, a "not affiliated with any government agency" disclaimer.

10. Analytics events (anonymous; PostHog)

- **v1 events kept:** onboarding, free test, question_answered (10% sampled), sim, paywall, purchase, restore, drill, glossary, readiness, trust, rating, app_open.
- **New events:**
- `onboarding_screen_viewed{n}`, `state_picker_repeat{count}` (loop detector)
- `text_size_changed{size}`, `sun_mode_toggled{on}`
- `proof_sheet_source_opened{qid}`, `official_manual_opened{state}`
- `replay_mistakes_started{n}`, `replay_mistakes_completed{n,correct}`
- `format_card_viewed{state}`
- `test_date_set{days_out}` (once Test Day ships)
- `cab_mode_started{section}`, `cab_mode_minutes`
- `paywall_viewed{trigger, section, after_free_test_completed:bool, mid_test:bool}`
- `fixlog_viewed`, `watch_flag_resolved{kind}` (server-side)
- **Guardrails** (alert if breached):
- `paywall_viewed{mid_test:true}` = **0**;
- `state_picker_repeat > 2` in < 1% of installs;
- mistake reports < 0.5 per 1,000 answers.
- **Gate dashboard:**
- store page→install; install→free test completed; free test→paywall→purchase;
- **paid sales/mo** (the gate metric), revenue per install;
- D1/D7 retention, report rate.

11. Milestones and acceptance (full v2, 21 days)

Milestone	Day	Accept when
M1 Foundations	3	App runs on iPhone and Android; content DB loads; question screen + autosave; CI lint; **design tokens (type ramp, color, Sun Mode)** in code; **snapshot test at the largest Dynamic Type size**; StateFormat rows for TX/CA/FL/NAT with sources
M2 Core loop	7	Practice with Proof sheet, simulator in state format, Results leading with Replay mistakes, drill, readiness, 3-screen onboarding with no-loop test, all offline on 100 sample Qs
M3 Content wave 1	10	GK, Air Brakes, Combination, HazMat at 70% of target; HazMat 100% human-checked; new lint clean
M4 Monetization	12	7 products sandbox-tested; Restore on a 2nd device; refunds handled; paywall rules pass QA (0 mid-test); no ads/tracking SDK
M5 Trust + accessibility	14	Source card on 100%; Report a mistake + offline queue; fix log; Cab Mode; VoiceOver/TalkBack; **3–5 person usability check done** (people Sean knows; no public recruiting posts without Sean's approval)
M6 Content complete	17	All sections at target; TX/CA/FL overrides; 20% sample error ≤ 2%
M7 Store-ready	19	Screenshots 1–6, listing, privacy labels, review notes, Maestro smoke on release builds, TestFlight + Play closed test live; rule-change watch running (if it fits)
M8 Submitted	21	Submitted to App Review and Play (or closed test running); Sean signs off

12. Schedules

12a. Full v2 schedule (21 days)

CC = Claude Code. Sean's time is 1–2 h/day plus review blocks.

Day	Claude Code	Sean
1	Scaffold, SQLite schema (§6, all 51 jurisdictions defined), CI; draft StateFormat TX/CA/FL/NAT with sources	Name + USPTO; Apple and Play accounts, Paid Apps, Small Business Program
2	Question screen, answer rows, autosave, source chip; lint script (v1 + new)	**Design day: tokens and type ramp in Figma (2h)**; verify StateFormat rows; download manuals with editions
3		Approve topic maps; create app records

Day	Claude Code	Sean
	Topic maps; start GK + HazMat drafts; tokens in code; snapshot test. **M1**	
4	Simulator + Results (Replay-first); choose RevenueCat or react-native-iap; draft Air Brakes	Create 7 IAPs; sandbox tester
5	**Proof sheet + Source card**; Replay drill; readiness; Home + format card; Check 1 on GK batch 1	HazMat batch 1 review (100%)
6	3-screen onboarding + no-loop Maestro test; interruptions/ resume; draft Combination + TX CR	**Design day: Proof sheet + Test Day wireframes**; start the Play closed test (R5)
7	Unit tests; demo build. **M2**	Device test; UX notes
8	HazMat → 115, GK → 175; Check 1	HazMat batch 2 (100%), GK sample
9	Air Brakes → 70, Combination → 55, TX CR → 40	TX CR (100%), sample
10	Integrate wave 1. **M3**	Sign off M3
11	IAP + strict paywall	Sandbox purchases
12	Restore, refunds, Play ack, SDK audit. **M4**	2nd-device restore
13	Report a mistake (Worker + D1), fix log; draft Tanker + Doubles	Support text; privacy policy
14	Cab Mode, glossary (80), **Sun Mode + accessibility pass**. **M5**	**Usability check (3–5 people)**; Cab Mode ride test
15	Passenger + School Bus; Check 1 all endorsements; TX/CA/FL overrides; **Test Day plan**	Endorsement samples
16	Fill to target; dedupe + verbatim; **Test Day plan finish**	Remaining HazMat + samples
17	Integrate; free sets. **M6**	Error-rate gate
18	Store assets (screenshots 1–6), listing, privacy; **rule-change watch job (eCFR + FR + 3 manual hashes)**	Approve name, icon, screenshots
19	Release builds; TestFlight / Play. **M7**	TestFlight run-through
20	Fixes; review notes; **5 SEO seed pages** (§13, Gate 0)	Final QA
21	Submit; runbook. **M8**	Submit; sign off

12b. Lean MVP: about 15 working days [E]

Cuts (all [LEAN-MOVE] items ship as OTA or v1.0.1 within days 22–35, no gate needed): 1. **Test Day plan + Ready check breakdown → v1.0.1.** Saves about 1.5 CC-days. Readiness on Home/Results stays. 2. **Endorsement banks at 50 each instead of 80; glossary 80 instead of 150.** Saves about 2 days of draft and review, and about 3h of Sean. 3. **Cab Mode background/lock-screen audio → v1.0.1.** Foreground Cab Mode stays. Saves about 0.5 day. 4. **Rule-change watch → days 22–35.** It's a server job with no app-review dependency. Saves about 1 day. The fix log screen still ships, fed by user reports. 5. **SEO seed pages → days 22–25.** Saves about 0.5 day. 6. **Screenshots 4–6 → after launch.** Launch with screenshots 1–3 plus 2 simple ones. Saves about 0.5 day. 7. **Android production can trail iOS** by the Play closed-test window (R5). The build is the same; this doesn't save days, it just means the launch isn't blocked on it.

What can't be cut (these are the differentiators): - exact-format engine for TX/CA/FL + NAT; - Proof sheet with citations; - the honest paywall; - Replay-first Results; - large type, Sun Mode and accessibility; - 3-screen no-loop onboarding; - HazMat + TX CR at 100% human-checked; - Report a mistake + fix log screen; - the usability check.

Lean day plan [E]:

Days	Work
1–3	Foundations (M1, design day 2)
4–6	Core loop + Proof sheet + onboarding (M2, design day 6)
7–9	Content wave 1 + IAP
10	Monetization (M4)

Days	Work
11	Trust + accessibility + Cab Mode (M5)
12	Usability check + content complete (M6 at lean targets)
13	Store assets
14	Release builds (M7)
15	Submit (M8)

- **Binding constraint:** Sean's review hours (about 10h total), not CC.
- **Days saved:** about 6 [E]. Under the factory model, that's about 6 days sooner to the day-30 and day-60 sales signals and to the next line in the water.
- **Recommendation [J]:** build the lean MVP. The cut items are polish or volume, and none of the three proof points depends on them.

13. Stage gates, revenue math and kill lines

Net per purchase (15% cut): Permit $12.74 · Endorsement $4.24 · All Endorsements $16.99 · Permit + All $29.73. $1k/mo net ≈ 54 sales/mo at an average ticket of about $22 [E]. Sales = paid purchases per 30 days, both stores.

Gate	Trigger	Unlocks (CC-days [E])	Sean
Gate 0: MVP launch	Submission (lean ≈ day 15)	App as §8. Within days 22–35, no gate: the [LEAN-MOVE] items, the **rule-change watch** (3–4), and **5 SEO seed pages** (`/hazmat-practice-test`, `/texas-cdl-practice-test`, `/cdl-air-brakes-practice-test`, `/texas-commercial-rules-test`, `/tanker-endorsement-practice-test`) built from the content DB (~1)	Search Console; triage reports
Day 30 check (early warning)	< 10 sales, or page→install < 20%, or install→purchase < 2%	No new scope. Rework listing, screenshots and paywall placement. Never add ads or subscriptions	Review the listing
Gate 1	**Day 30 ≥ 10 sales and days 31–60 ≥ 25 sales/mo**	Programmatic **SEO pages** per state × test (4–6); **all 51 jurisdictions** in StateFormat with format cards (6–8 CC-days + about 17h of human verification at 20 min/row [E]; NY/IA 3-choice etc.); **endorsement modes**: HazMat renewal, air-brake L/Z restriction removal, CLP→skills checklist (4–5); endorsement banks to 80	State-row verification (or Lindsey)
Kill line (day 60)	Days 31–60 **< 25 sales/mo and not rising week over week**	**Stop new feature work.** Maintenance mode (§13.1). Rotate the content engine to the next exam (§16)	Moonbase check on the next niche
Gate 2	**≥ 50 sales/mo for 4 straight weeks**	**Miss-rate pages** (opt-in aggregate data; "hardest 25 questions in Texas") (3); **Spanish English-test bridge** (bilingual toggle, English term drill, Roadside English, "your state tests in English" banner) (5–7 + paid fluent reviewer, which is a purchase needing Sean's OK); **pre-trip audio trainer** (4–5); **video pipeline** (Q&A videos from the bank, EN + ES, about 2/wk, no paid spend) (3); test-day topic check-in (topics only, never questions) (2)	Spanish review option; video voice approval
Gate 2 hold	25–49 sales/mo	Keep Gate 1 work going for 30 more days, then re-check. If still < 25 at day 90, treat it as the kill line	—
Gate 3	**≥ 80 sales/mo for 4 straight weeks**	**Voice answers** in Cab Mode (3); **price test**: Permit Pack $14.99 → $19.99 for 2 weeks, then compare sales × net; **motorcycle extension** reusing the engine (6,600 searches/mo, KD 30) as a separate app per §16; **optional paid examiner review** ("Reviewed by a retired CDL examiner"), a one-off purchase that **needs Sean's explicit OK** (price not researched)	Approve any spend

Rules: - A gate unlocks work; it never forces it. - Lindsey or agent time freed by a kill goes to the next line. - Gates are checked weekly in the factory dashboard.

13.1 Maintenance mode (after a kill, or between gates)

- The app stays live; it costs nothing.

- The rule-change watch keeps running (free tier). Reports are triaged weekly, not every 72h.
- Fixes ship only for wrong answers or format changes.
- About 1h/week of Sean, maximum [E].
- If sales later cross 25/mo organically, re-enter at Gate 1.

14. Risks (v1 R1–R9 kept; new and changed)

#	Risk	Mitigation
R1–R9	As v1	As v1
R4 (updated)	State format or rule changes	The rule-change watch (§7.8) + StateFormat source/last_checked + public fix log
R10	A polished no-ads rival (Hanh Le cheaper lifetime; indie "CDL Prep Ultimate")	Lead with exactness and proof, not design alone; the fix-log record compounds
R11	A source excerpt drifts toward manual text	≤ 240-char paraphrase rule + verbatim CI check
R12	Lean banks feel thin (50-Q endorsements repeat)	The simulator draws without repeats until the bank is exhausted; banks grow OTA at Gate 1; state the counts honestly in the listing
R13	**Factory 4.3 risk:** many template-based exam apps from one developer account read as spam	§16 rules: distinct brand, UI theme, content and review notes per exam; never per-state clones; space submissions
R14	Spanish/immigrant demand shrinks (FL/TX English-only; non-domiciled rule)	Spanish waits for Gate 2 and is framed as a bridge to the English test

15. Review checklist before submission (new, short)

- [ ] 0 mid-test paywall views across the Maestro suite
- [ ] No truncation at the largest Dynamic Type size (snapshot)
- [ ] Onboarding no-loop test green on 4 sizes
- [ ] 100% of questions have a citation, Source card and Last checked
- [ ] HazMat + TX CR 100% human-checked; sample error ≤ 2%
- [ ] StateFormat TX/CA/FL/NAT verified with sources
- [ ] Restore on a fresh install and a 2nd device
- [ ] No ads/attribution/tracking SDK; privacy labels match
- [ ] Disclaimer and no pass claims

16. Factory template (reuse for the next licensing-exam apps)

Each module is built once here, then reused. Each new exam app changes the content and brand, not the machinery.

Module	What's reusable as-is	Per-exam work [E]
Content engine (schema §6, pipeline §7, lint, two-source + human gate, OTA updates)	Question/Source/FixLog types, agent draft → Check 1 → human gate, all lint rules, verbatim check, dedupe	New `Section` list, topic maps, sources; banks drafted (CC) and reviewed (Sean)
Exact-format engine (StateFormat-style table + format card + simulator)	Jurisdiction × test table with source/last_checked; the simulator reads it	Rows for that exam (some are federal-only, e.g. one national format)
Rule-change watch + public fix log	eCFR/Federal Register/URL-hash jobs, flag propagation, `/fixes` page	Point at that exam's CFR parts, agencies and manuals
Honest-paywall module	One-time IAP wrapper, Restore, entitlement cache, §4 rules, guardrail metric, bundle math	SKU list and prices
Proof-sheet UI kit	Question screen, Proof sheet, Source card, Results with Replay, Replay drill, readiness, Cab Mode, Sun Mode, large type, a11y, 3-screen onboarding + no-loop test	**New theme per app** (color, icon, type pairing) to avoid 4.3
SEO page generator (Astro)	Builds topic and jurisdiction pages, a free quiz widget and Quiz/FAQ schema from the content DB	Keyword list (≤ 15 Semrush rows per niche)
Stage gates + kill lines		

Module	What's reusable as-is	Per-exam work [E]
	Gate 0–3 structure, day-30 warning, day-60 kill (< 25/mo), maintenance mode, gate dashboard	Thresholds scaled to price (sales needed for $1k/mo net)
Analytics and launch checklist	Event schema, guardrails, §15 checklist, review-notes template	Exam-specific events

Rules for reuse: 1. **One app per exam.** Never per-state or per-endorsement clones. 2. A **distinct brand, theme and screenshots** per app. Review notes explain the original content and sources (4.3). 3. A **Moonbase check** before each new niche. 4. **No services, no paid ads, no outreach**, the same as here.

Estimated cost of app #2 [E]: - Ship in about 8–12 working days, versus about 15 lean here. The saving is mostly UI, IAP, CI and pipeline reuse. - Content review stays the bottleneck, scaling with bank size. - Candidates are already in the queue (e.g. Part 107 is #1). Motorcycle reuses this engine at CDL Gate 3. A car permit app only after a winner, because that market is crowded (33k/mo, KD 59).